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DATA SHEET

G2 FT2232H PORT
Revision 1.0

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FT2232H MPSSE Port for the G2 Amiga: JTAG, SPI Flash, GPIO and STM32 SWD

An FTDI FT2232H on a port of the G2's USB hub, run by the firmware's EHCI host, whose channel A in MPSSE mode is reached from AmigaOS as a byte stream through `g2serial.device`. Three AmigaOS programs use it: ftdimon (the panel: device, pins, JTAG, SPI flash, hex editor), jtagtool (loopback, chain scan, Xilinx configuration) and swdtool (ARM Serial Wire Debug for STM32 parts). Figures are measured on G2 unit A; functions not yet run on hardware are marked untested.

HIGHLIGHTS

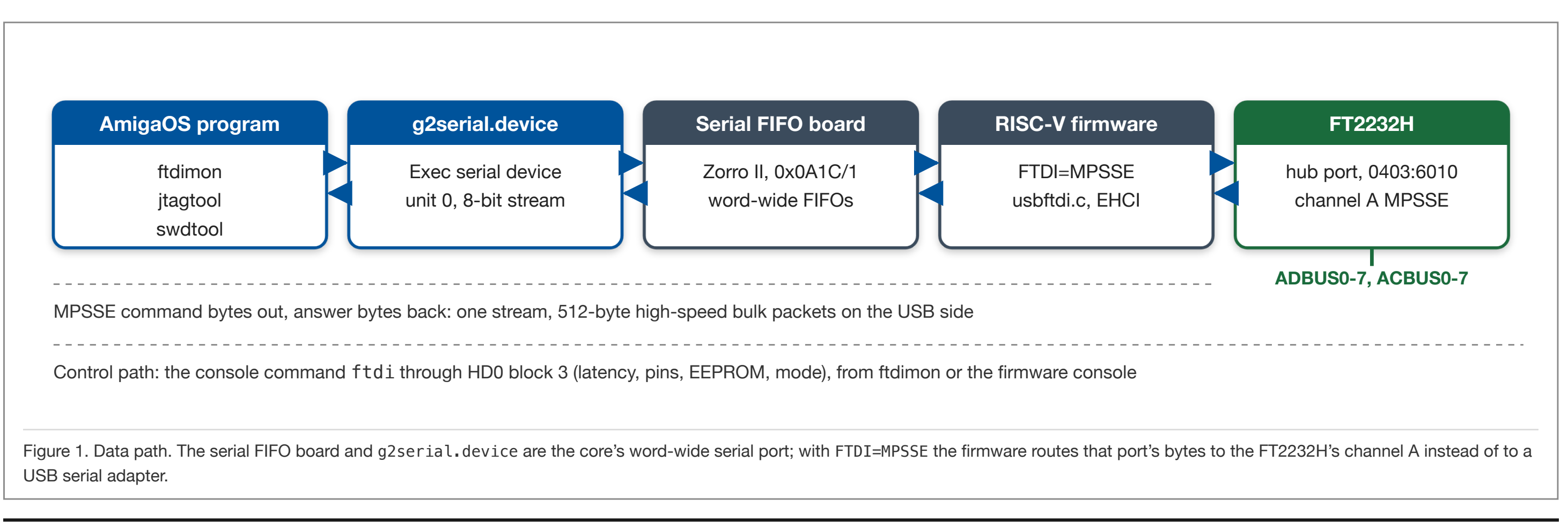
- **Peripheral:** any FT2232H breakout (0403:6010) on a G2 hub port; channel A opened by the firmware with 512-byte high-speed bulk packets and put in MPSSE mode by `FTDI=MPSSE` in `\amiga\AMIGA.INI` (menu item FTDI).
- **Stream:** MPSSE command bytes (FTDI AN_108) written to `g2serial.device`; answers read back from it. Loopback through the chip: 0 bit errors in 1,024,000 bytes at TCK 1, 6 and 30 MHz; about 14,000 bytes/s each way.
- **Control:** the firmware command `ftdi` (latency timer, pins, raw vendor requests, EEPROM read/write/erase, MPSSE or serial mode) through HD0's command block.
- **SPI NOR flash:** JEDEC ID, 402-part table from flashrom 1.4.0, read, erase, program, verify; SPI modes 0–3, MSB or LSB first, chip select on ADBUS3–7. AT25SF081 1 MB written and verified, 0 bytes differ.
- **JTAG:** chain scan (IDCODE, vendor, part, IR length); configuration of an XC3S1600E or XC6SLX100 from a `.bit` file and JPROGRAM reboot .
- **STM32 SWD:** ADIv5 SW-DP and MEM-AP; device identification of 46 STM32 IDs; memory read/write; flash erase, program and verify on the F0/F1/F3 and F2/F4/F7 flash interfaces; reset and halt.
- **GPIO:** all 16 pins of channel A as inputs or outputs, latched and read ten times a second.

KEY PARAMETERS

PARAMETER	VALUE
Chip	FTDI FT2232H, USB 2.0 high speed, 0403:6010
Channel used	`A (interface 0, wIndex 1)`
Mode	`MPSSE (SET_BITMODE 0x02)`
I/O level	`3.3 V (inputs 5 V tolerant)`
Clock (TCK, SCK, SWCLK)	`30 MHz / (1 + divisor): 458 Hz–30 MHz`
USB packets	`512 B bulk IN and OUT`
Stream rate (measured)	`13,000–14,300 bytes/s each way`
SPI flash read (measured)	`12,692 bytes/s (1 MB in 1:22)`
Amiga driver	`g2serial.device unit 0`
Firmware switch	`AMIGA.INI FTDI=MPSSE | SERIAL`
Programs	`ftdimon, jtagtool, swdtool`
Source	`amiga/tools/ftdimon, amiga/tools/jtag, fw/usbftdi.c`

1 FUNCTION

The FT2232H holds two channels; each can be a UART, a FIFO, a bit-bang port or a Multi-Protocol Synchronous Serial Engine (MPSSE). The MPSSE clocks bytes and bits in and out on ADBUS0–2 with a programmable clock, drives ADBUS3 as TMS, and sets and reads the remaining pins of the channel as general-purpose I/O. Every operation is a command byte followed by its parameters; read commands return answer bytes. The G2 Amiga places channel A in this mode and passes its command and answer bytes unchanged between the chip and an AmigaOS serial device, so any MPSSE protocol is written on the Amiga as a stream of bytes.



MPSSE command bytes out, answer bytes back: one stream, 512-byte high-speed bulk packets on the USB side

Control path: the console command `ftdi` through HD0 block 3 (latency, pins, EEPROM, mode), from ftdimon or the firmware console

Figure 1. Data path. The serial FIFO board and `g2serial.device` are the core's word-wide serial port; with `FTDI=MPSSE` the firmware routes that port's bytes to the FT2232H's channel A instead of to a USB serial adapter.

BLOCK	IMPLEMENTATION	SOURCE
AmigaOS programs	Build MPSSE command streams, read answers, run the protocols (JTAG TAP, SPI NOR, SWD)	`amiga/tools/ftdimon`, `amiga/tools/jtag`
MPSSE engine	Command queue, answer collection into bit positions (AN_108 3.3: a bit read lands in the top bits of its byte), sync by the bad command 0xAA (answer 0xFA 0xAA)	`amiga/tools/jtag/mpsse.c`
Serial stream	Exec serial device opened at 8 bits, reads with a timeout	`amiga/tools/xfer/zio_amiga.c`, `g2serial.device`
Firmware FTDI driver	Opens the first FTDI device (vendor 0x0403) on the hub; a multi-channel chip on channel A; 512-byte packets in DMA RAM windows 0x7400 (IN) and 0x7600 (OUT); strips the two status bytes of every IN packet; SET_BITMODE 0x02 after both FIFOs are purged	`fw/usbftdi.c`, `fw/amiga/amiga_main.c`
USB host	EHCI 1.0 controller and the USB2514 hub of the G2	`fw/usb_ehci.c`

2 SUPPORTED FUNCTIONS

FUNCTION	PROGRAM	OPERATIONS	STATE
Device	ftdimon page 1	Chip state from the firmware; clock (30 MHz to 100 kHz), divide by 5, adaptive and three-phase clocking; internal loopback; latency timer; channel mode; EEPROM read, write, erase	Tested; EEPROM write and erase untested (no EEPROM fitted on the breakout)
GPIO	ftdimon page 2	ADBUS0–7, ACBUS0–7: direction and level set (0x80, 0x82), read (0x81, 0x83) ten times a second	Tested
JTAG chain	ftdimon page 3, jtagtool SCAN	TAP reset; IDCODE, vendor, part and IR length of each device	Tested on a wire (0 devices); untested on a real chain
JTAG loopback	ftdimon page 3, jtagtool LOOP	Internal (0x84) and external (ADBUS1 to ADBUS2) rounds of a pattern; bit errors, bytes/s	Tested: 0 bit errors
Xilinx configuration	jtagtool PROGRAM, REBOOT	XC3S1600E (Pano G1) or XC6SLX100 (Pano G2 revision C) from a .bit file: IDCODE check, JPROGRAM, CFG_IN, JSTART, DONE; JPROGRAM to reload from flash	Untested on this engine (the sequence is panoflash's, proven on a Platform Cable)
SPI NOR flash	ftdimon page 4	Detect (0x9F), unprotect, read, erase (by the part's erase blocks or chip), program (256-byte pages, blank pages skipped), verify, Write All; CRC32 of file and chip	Tested on an AT25SF081
Hex editor	ftdimon page 5	Flash buffer (up to the part size) and EEPROM 512 bytes; edit, go to, open, save, fill	Tested
STM32 SWD	swdtool	ID, READ, WRITE (erase, program, verify, reset), ERASE (all or range), RESET (HALT), PEEK, POKE	New; untested on hardware (no STM32 wired)

3 PIN ASSIGNMENT

All functions use channel A. Pin numbers are those of the FT2232H package (LQFP-64 and QFN-64 share the numbering, FTDI DS_FT2232H); a breakout board labels the same signals AD0–AD7 and AC0–AC7 on its headers. Only one function is wired at a time: the programs set the pin directions they need when they start.

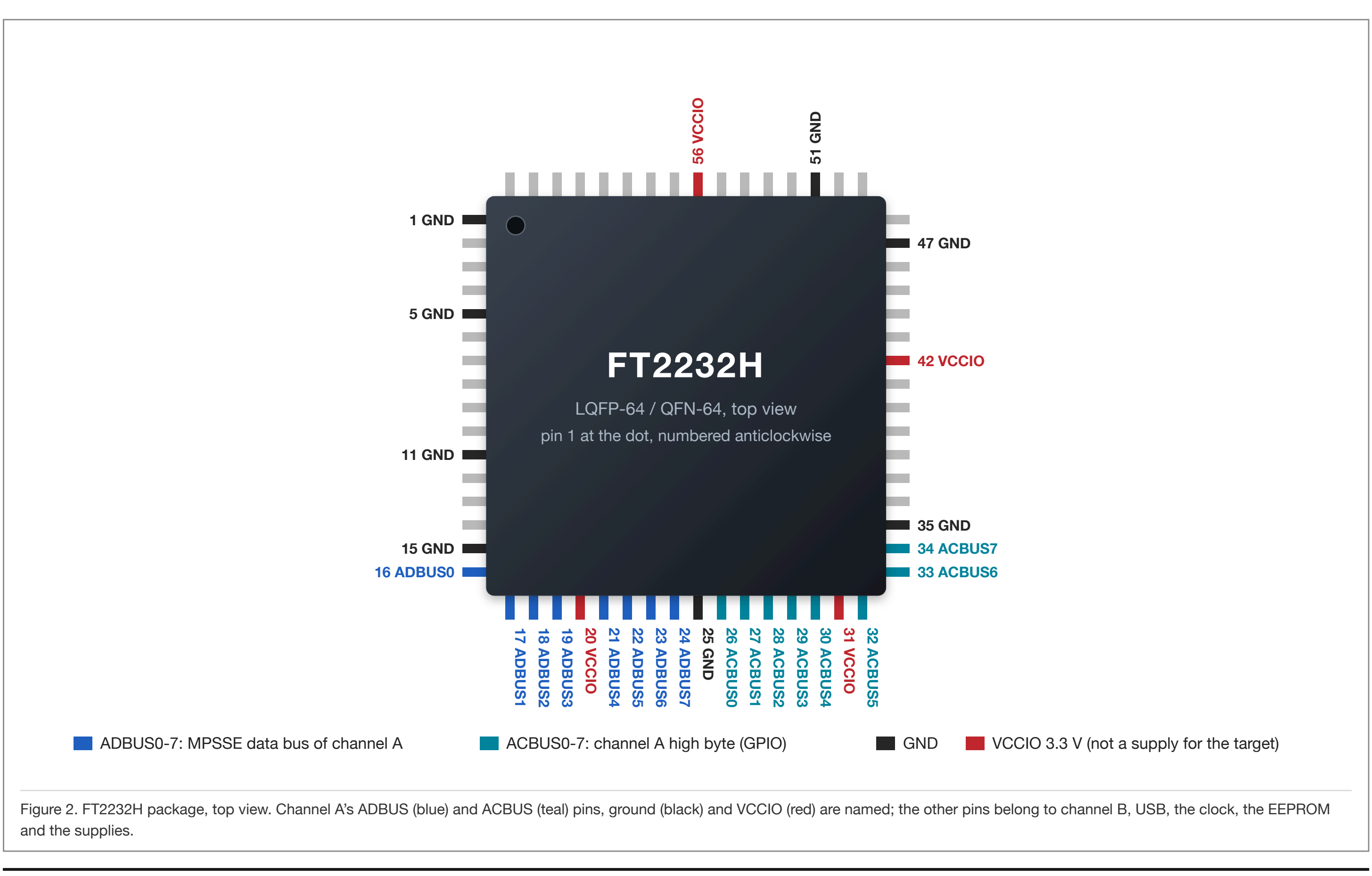


Figure 2. FT2232H package, top view. Channel A's ADBUS (blue) and ACBUS (teal) pins, ground (black) and VCCIO (red) are named; the other pins belong to channel B, USB, the clock, the EEPROM and the supplies.

PIN	SIGNAL	JTAG	SPI FLASH	STM32 SWD	PINS PAGE
16	ADBUS0	TCK	SCK	SWCLK	In/Out, Lo/Hi
17	ADBUS1	TDI	MOSI	SWDIO through 470 Ω	In/Out, Lo/Hi
18	ADBUS2	TDO	MISO	SWDIO (read)	In/Out, Lo/Hi
19	ADBUS3	TMS	CS# when CSPin=3	–	In/Out, Lo/Hi
21	ADBUS4	–	CS# (default)	NRST (optional)	In/Out, Lo/Hi
22	ADBUS5	–	CS# when CSPin=5	–	In/Out, Lo/Hi
23	ADBUS6	–	CS# when CSPin=6	–	In/Out, Lo/Hi
24	ADBUS7	–	CRESET (iCE40), or CS# when CSPin=7	–	In/Out, Lo/Hi
26–30, 32–34	ACBUS0–7	–	–	–	In/Out, Lo/Hi
1, 5, 11, 15, 25, 35, 47, 51	GND	GND	GND	GND	
20, 31, 42, 56	VCCIO	3.3 V I/O supply of the chip			

▬ clock ▬ data out of the FT2232H ▬ data into the FT2232H ▬ select / mode ▬ reset or supply ▬ ground

4 WIRING

4.1 SPI NOR flash (25-series, SOIC-8)

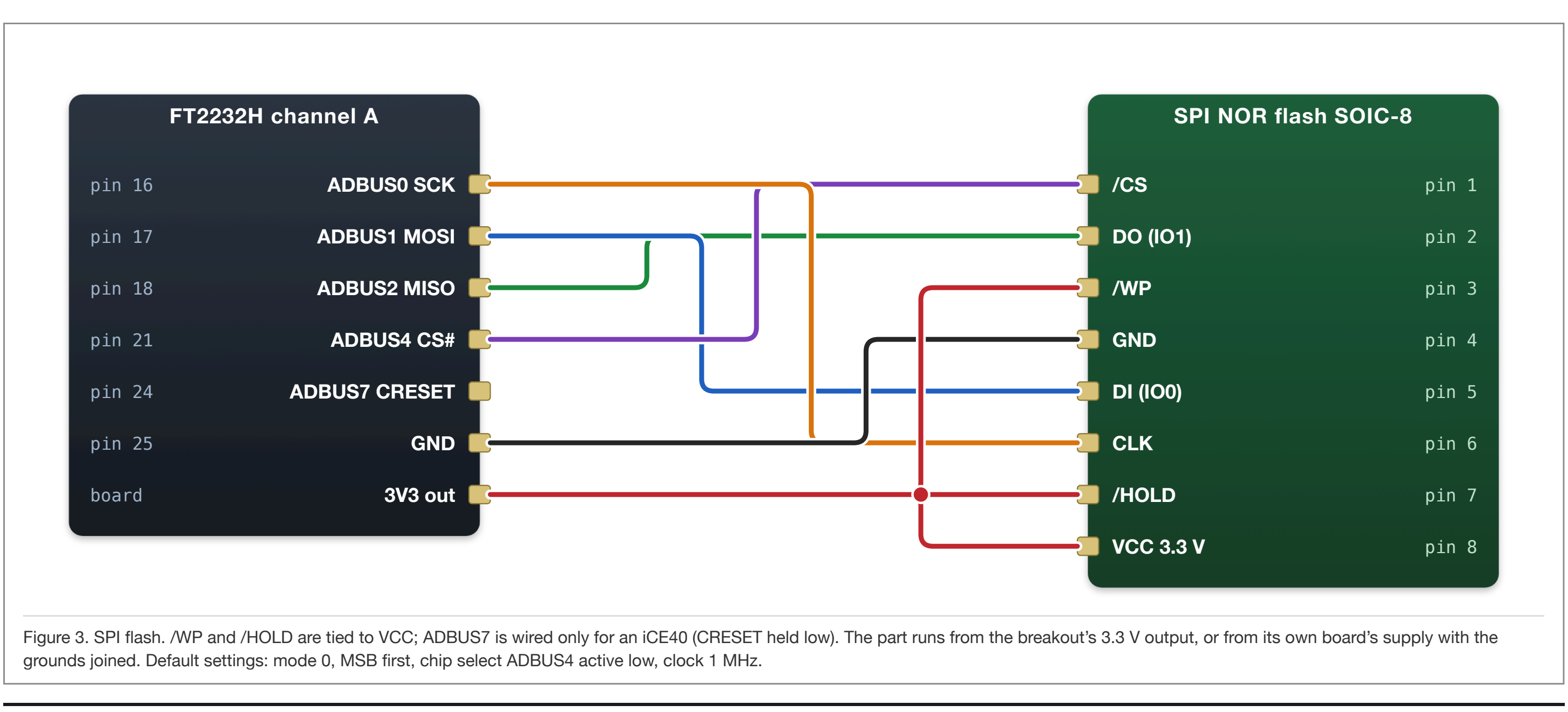


Figure 3. SPI flash. /WP and /HOLD are tied to VCC; ADBUS7 is wired only for an iCE40 (CRESET held low). The part runs from the breakout's 3.3 V output, or from its own board's supply with the grounds joined. Default settings: mode 0, MSB first, chip select ADBUS4 active low, clock 1 MHz.

FT2232H PIN	SIGNAL	FLASH PIN (SOIC-8)	NOTE
16 ADBUS0	SCK	6 CLK	
17 ADBUS1	MOSI	5 DI (IO0)	
18 ADBUS2	MISO	2 DO (IO1)	
21 ADBUS4	CS#	1 /CS	CSPin=4, CSHigh=0 (ftdimon Preferences)
24 ADBUS7	CRESET	–	An iCE40 FPGA's CRESET_B when its flash is programmed in circuit; held low during the job
GND	GND	4 GND	
3V3 of the breakout	VCC	8 VCC, 3 /WP, 7 /HOLD	/WP and /HOLD high: no write protect, no hold

4.2 JTAG (Xilinx 2 × 7 header, 2 mm)

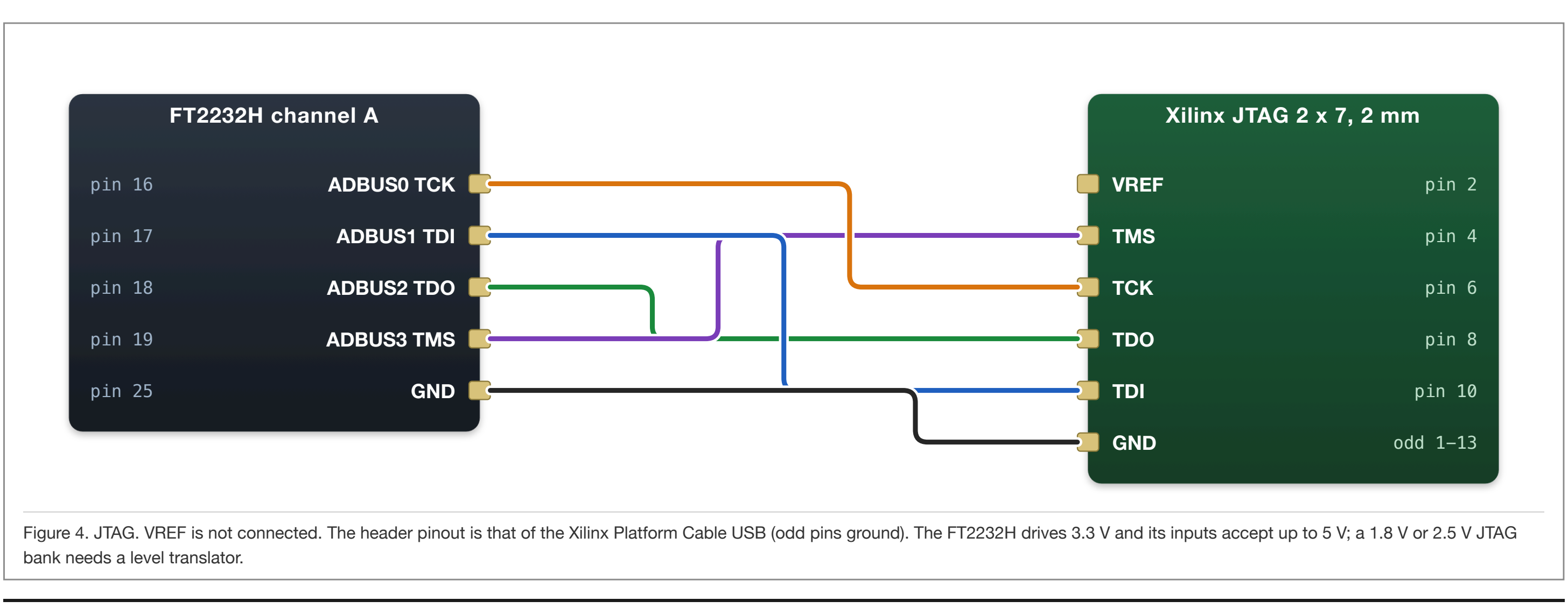


Figure 4. JTAG. VREF is not connected. The header pinout is that of the Xilinx Platform Cable USB (odd pins ground). The FT2232H drives 3.3 V and its inputs accept up to 5 V; a 1.8 V or 2.5 V JTAG bank needs a level translator.

FT2232H PIN	SIGNAL	XILINX 14-PIN HEADER	DIRECTION
16 ADBUS0	TCK	6	out
17 ADBUS1	TDI	10	out
18 ADBUS2	TDO	8	in
19 ADBUS3	TMS	4	out
GND	GND	1, 3, 5, 7, 9, 11, 13	
–	VREF	2	not connected

4.3 STM32 Serial Wire Debug

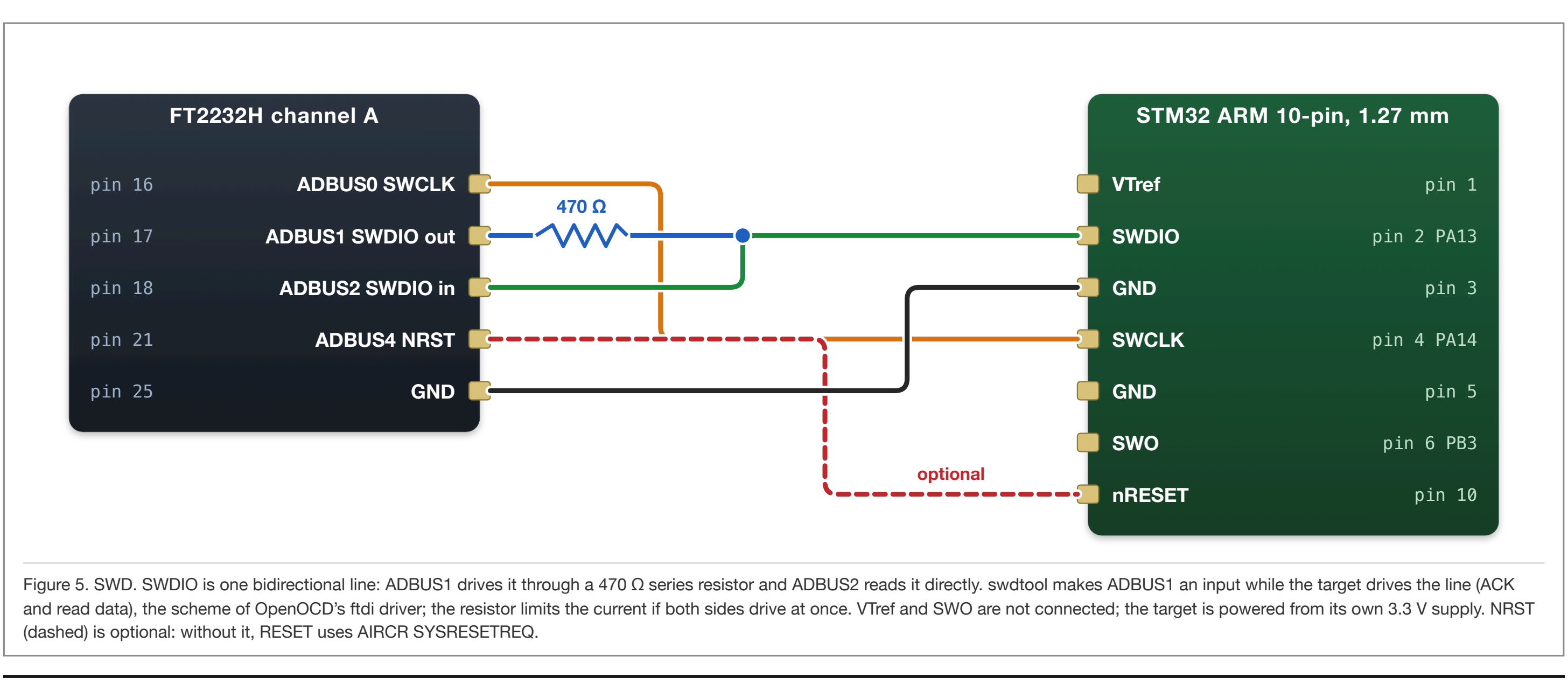


Figure 5. SWD. SWDIO is one bidirectional line: ADBUS1 drives it through a 470 Ω series resistor and ADBUS2 reads it directly. swdtool makes ADBUS1 an input while the target drives the line (ACK and read data), the scheme of OpenOCD's ftdi driver; the resistor limits the current if both sides drive at once. VTref and SWO are not connected; the target is powered from its own 3.3 V supply. NRST (dashed) is optional: without it, RESET uses AIRCR SYSRESETREQ.

FT2232H PIN	SIGNAL	ARM 10-PIN (1.27 MM)	ST-LINK 20-PIN	STM32 PIN
16 ADBUS0	SWCLK	4	9	PA14
17 ADBUS1	SWDIO, through 470 Ω	2	7	PA13
18 ADBUS2	SWDIO, direct	2	7	PA13
21 ADBUS4	NRST (optional)	10	15	NRST
GND	GND	3, 5, 9	4, 6, 8 … 20	VSS
–	VTref	1	1	not connected (3.3 V targets only)

On a board with a 4-pin SWD header (3V3, SWDIO, SWCLK, GND), the order is printed on the board; connect by name. The target is powered from its own supply.

4.4 Loopback test

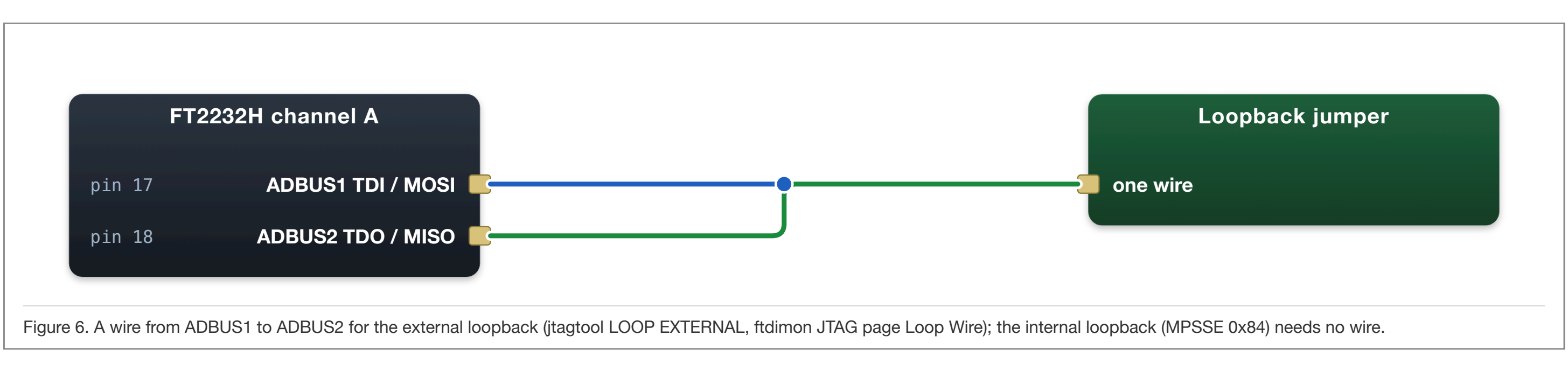


Figure 6. A wire from ADBUS1 to ADBUS2 for the external loopback (jtagtool LOOP EXTERNAL, ftdimon JTAG page Loop Wire); the internal loopback (MPSSE 0x84) needs no wire.

5 SOFTWARE INTERFACE

5.1 Enabling the port

Set `FTDI=MPSSE` in `\amiga\AMIGA.INI` or choose MPSSE at the setup item FTDI channel A (Peripherals page); the firmware applies it at once and on every start. With `FTDI=SERIAL` (the default) an FTDI chip is a USB serial adapter behind the same driver. The firmware console line `FTDI:` of `status`, or `ftdi`, shows the state.

5.2 Firmware command `ftdi`

Run at the firmware console, or from AmigaOS through HD0's command block (ftdimon's `link.c fw_cmd`). Every answer starts `ftdi: ok` or `ftdi: error`.

COMMAND	EFFECT	ANSWER
`ftdi`	State	`ftdi: ok port P vid 0403 pid 6010 index 1 iface 0 mode mpsse packets 512 latency L pins XX in N out N errors I/O dropped D`
`ftdi mode mpsse|serial`	Channel mode; written to AMIGA.INI	`ftdi: ok`
`ftdi ctl TT:RR VVVV [IIII|-] [len]`	A control request: bmRequestType TT, bRequest RR, wValue, wIndex (default: the channel), up to 64 bytes read	`ftdi: ok` and the bytes in hex
`ftdi ee`	EEPROM read, 256 words (93C66; a 93C46/56 repeats)	16 words a line
`ftdi ee write AA VVVV`	One EEPROM word	`ftdi: ok`
`ftdi ee erase`	The whole EEPROM	`ftdi: ok`

5.3 MPSSE commands used (FTDI AN_108)

COMMAND	USE
`0x19 / 0x1B`	Bytes / 1–8 bits out on the falling edge, LSB first (TDI, MOSI, SWDIO request and write data)
`0x39 / 0x3B`	Bytes / bits out and in (TDI and TDO together)
`0x28 / 0x2A`	Bytes / bits in on the rising edge, LSB first (SWD ACK and read data)
`0x11, 0x20, 0x31 / 0x10, 0x24, 0x34`	SPI MSB-first transfers, modes 0 and 3 / modes 1 and 2
`0x4B / 0x6B`	TMS bits (and TDO read): TAP navigation
`0x80, 0x81 / 0x82, 0x83`	ADBUS / ACBUS value and direction; read
`0x84 / 0x85`	Internal loopback on / off
`0x86`	Clock divisor: f = 30 MHz / (1 + divisor) with 0x8A (divide by 5 off)
`0x87`	Send the answers now
`0x8A/0x8B, 0x8C/0x8D, 0x96/0x97`	Divide by 5, three-phase clocking, adaptive clocking
`0x8F`	Clock without data (SWD idle cycles)
`0xAA`	Bad command: answered 0xFA 0xAA; the stream sync

5.4 ftdimon

A window in the house style with five pages (keys 1–5): Device, Pins, JTAG, SPI Flash, Hex Editor. Preferences (`,` or Project menu) are saved as `ENV:ftdimon/` variables: SCKIdle, SampleEdge, LSBFirst, CSPin (3–7, default 4), CSHigh, CReset (0 low, 1 high, 2 not driven), Clock (0–7, default 5 = 1 MHz), Divide5, Adaptive, ThreePhase, Latency (1–255 ms), StartPage. Details: `amiga/tools/ftdimon/README.md`.

5.5 jtagtool

```
jtagtool [DEVICE=g2serial.device] [UNIT=n] [CLOCK=hz] LOOP [INTERNAL|EXTERNAL|BOTH] [BYTES=1–4096] [ROUNDS=n]
jtagtool [DEVICE=...] [CLOCK=hz] SCAN
jtagtool [DEVICE=...] [CLOCK=hz] PROGRAM FILE=design.bit
jtagtool [DEVICE=...] [CLOCK=hz] REBOOT
```

CLOCK default 1,000,000 Hz. PROGRAM loads the FPGA only (the configuration is lost at power-off); it refuses an IDCODE other than XC3S1600E (0x01C3A093) or XC6SLX100 (0x04011093). Return codes: 0 pass, 5 fail, 10 usage, 20 no stream or no MPSSE sync.

5.6 swdtool

```
swdtool [DEVICE=g2serial.device] [UNIT=n] [CLOCK=hz] [NRST] ID
swdtool ... READ FILE=f [ADDR=a] [LEN=n]                memory to a file; default the whole flash
swdtool ... WRITE FILE=f [ADDR=a] [NOERASE] [NOVERIFY] [NORUN]
swdtool ... ERASE ALL | ERASE ADDR=a LEN=n
swdtool ... RESET [HALT]
swdtool ... PEEK ADDR=a [LEN=n] | POKE ADDR=a VALUE=v
```

ITEM	BEHAVIOUR
Connect	56 cycles high, JTAG-to-SWD sequence 0xE79E, 56 cycles high, idle; DPIDR; ABORT; CTRL/STAT CSYSPWRUPREQ, CDBGPWRUPREQ and ORUNDETECT; MEM-AP 0 CSW 32-bit with auto-increment
Transactions	Queued and sent as one MPSSE exchange, up to 120 words; with overrun detection a WAIT or FAULT keeps its data phase, so the stream stays aligned; reads are retried after a line reset and ABORT
ID	DPIDR, AP IDR, CPUID (core and revision), DBGMCU_IDCODE (0xE0042000; 0x40015800 on Cortex-M0/M0+; 0x5C001000 on STM32H7), device and revision, flash size
Flash, F0/F1/F3	Interface at 0x40022000: unlock (KEYR 0x45670123, 0xCDEF89AB), page erase (PER, AR, STRT), mass erase (MER), halfword programming (PG) with 75 µs of idle cycles after each; XL-density bank 2 above 0x08080000. RM0008, RM0091, RM0316
Flash, F2/F4/F7	Interface at 0x40023C00: unlock, sector erase (SER, SNB, STRT), mass erase (MER, MER1 on dual-bank 2 MB F42x/43x), word programming with PSIZE x32 (needs VDD 2.7–3.6 V) and 30 µs idle after each. RM0033, RM0090, RM0385, RM0410, RM0431
Other families	H7, L0, L1, L4, G0, G4, WB: identified; memory read and written; flash not programmed
Read protection	A flash read under RDP level 1 ends in a bus fault and is reported; level 1 is not removed by swdtool
Return codes	0 done, 5 failed, 10 usage, 20 no stream or no answer from the target

5.7 C interface

HEADER	FUNCTIONS
`xpc.h`	`mpsse_open, mpsse_cmd, mpsse_xfer, mpsse_loopback, xpc_tms, xpc_shift, xpc_flush`
`jtag.h, chain.h`	`jtag_init, jtag_reset, jtag_shift_ir, jtag_shift_dr, jtag_idle, chain_scan`
`s3e.h, bitfile.h`	`s3e_idcode, s3e_usercode, s3e_program, s3e_reboot, s3e_done, bitfile_load, bitfile_parse`
`swd.h`	`swd_open, swd_connect, swd_line_reset, swd_q, swd_run, swd_clear, swd_dp_read/write, swd_ap_read/write, swd_read32, swd_write32, swd_read_block, swd_write_block, swd_write_half, swd_nrst`
`spi.h` (ftdimon)	SPI NOR procedures of ftdiflash on the MPSSE stream; `spi_config`

6 MEASURED RESULTS (G2 REVISION C UNIT)

TEST	RESULT
Device page, FT2232H breakout on hub port 4	0403:6010, channel A, mpsse, 512-byte packets, latency 1 ms, pins F8
Internal loopback, jtagtool, 100 × 1,024 bytes at 1 MHz	0 bit errors, 13,008 bytes/s, longest round trip 88 ms
Internal loopback, 50 × 4,096 bytes at 6 and 30 MHz	0 bit errors, 14,288 and 14,253 bytes/s
External loopback ADBUS1–ADBUS2, 1, 6, 30 MHz	0 bit errors; 13,018, 14,295, 14,271 bytes/s
External loopback, no wire	23,552 bit errors, TDO read 1: fail as expected
ftdimon Loop Wire, 20 × 1,024 bytes	0 bit errors, 13,864 bytes/s
SPI detect, AT25SF081	ID 1F 85 01, 1,024 KB, page 256 bytes, erasers 20 52 D8 60 C7
SPI mode against clock, 0x9F	Mode 0 correct at 1, 6, 30 MHz; mode 3 correct at 6 MHz only
SPI read 1,048,576 bytes	1:22, 12,692 bytes/s; equal to the same chip read by ftdiflash on a Mac
SPI Write All 0x000000–0x0FFFFF	Erase about 15 s, program, verify 1:29: 0 bytes differ
CRC32 (68020 assembler) against zlib	Equal; 1,048,576 bytes in 841 ms

The rate does not depend on the clock: the path through `g2serial.device`, the serial FIFO board and the firmware carries about 14,000 bytes/s each way. Estimates from that rate (untested): a 723 KB Xilinx G1 configuration about 50 s; swdtool reads about 3,000 bytes/s (19 command bytes per 32-bit word) and programs an F1 at about 1,100 bytes/s (27 command bytes per halfword).

7 LIMITS AND UNTESTED FUNCTIONS

ITEM	STATE
STM32 SWD (swdtool)	Untested: no STM32 wired to the test unit; built with 0 warnings, installed and run up to the target connection
Xilinx configuration (jtagtool PROGRAM, REBOOT)	Untested on the MPSSE engine
JTAG scan of a real chain	Untested: no target wired
EEPROM write and erase	Untested: the breakout has no EEPROM fitted
SWD on Cortex-M targets other than STM32	ID, PEEK, POKE, READ and RESET are generic ADIv5; flash programming is STM32 only
Voltage	3.3 V signalling only; no VTref sense, no level translation
Channel B	Not opened by the firmware (FT2232H_PORT_SPEC.md sections 1, 5, 6)
ftdimon SWD page	Not present; SWD is the command swdtool
Other screen modes of ftdimon (PAL 640 × 256, interlace)	Untested

8 COMPONENTS AND LICENCES

COMPONENT	AUTHOR	REVISION	LICENCE
ftdimon, jtagtool, swdtool, mpsse.c, swd.c, link.c, firmware FTDI driver and command	Mike Wolak, EPROM Foundry Inc., pano-g2-amiga	release 1.0	GPL-2.0-or-later
jtag.c, chain.c, s3e.c, bitfile.c (TAP, chain scan, Xilinx configuration)	Mike Wolak, EPROM Foundry Inc., the G1/G2 JTAG flasher	unchanged copy	project
ftdiflash (SPI NOR procedures)	Clifford Wolf (iceprog, Project IceStorm); Dean Miller, Adafruit Industries	242ae956	ISC
flashrom chip table (402 SPI NOR parts)	flashrom project	1.4.0	GPL-2.0-or-later
FTDI requests and status bytes	TinyUSB host serial class (hathach); FTDI AN232B-05	–	MIT
SWD pin scheme and sampling edges	OpenOCD ftdi driver (reference only, no code copied)	–	GPL-2.0
FT2232H, MPSSE	Future Technology Devices International: DS_FT2232H, AN_108	–	data sheet
ADIv5 SW-DP, MEM-AP	Arm Ltd: IHI 0031 (ARM Debug Interface Architecture Specification)	ADIv5	specification
STM32 flash interfaces	STMicroelectronics: RM0008, RM0033, RM0090, RM0091, RM0316, RM0385, RM0410, RM0431	–	reference manuals

Measured on a Pano Logic G2 revision C unit with jtagtool and ftdimon (fabric 0x0AB5EF2D). FT2232H pin numbers from FTDI DS_FT2232H (LQFP-64/QFN-64). Repository `mikewolak/pano-g2-amiga`, release 1.0.